

Avances2.6

Andre Jo 22199

2025-04-25

```
datos <-read.csv("test.csv")
```

```
##      Id SalePrice economico medio caro
## 1 1461  169277.1         1      0     0
## 2 1462  187758.4         0      1     0
## 3 1463  183583.7         1      0     0
## 4 1464  179317.5         1      0     0
## 5 1465  150730.1         1      0     0
## 6 1466  177151.0         1      0     0
```

##	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour
## 1	1	60	RL	65	8450	Pave	<NA>	Reg	Lvl
## 2	2	20	RL	80	9600	Pave	<NA>	Reg	Lvl
## 3	3	60	RL	68	11250	Pave	<NA>	IR1	Lvl
## 4	4	70	RL	60	9550	Pave	<NA>	IR1	Lvl
## 5	5	60	RL	84	14260	Pave	<NA>	IR1	Lvl
## 6	6	50	RL	85	14115	Pave	<NA>	IR1	Lvl
##	Utilities	LotConfig	LandSlope	Neighborhood	Condition1	Condition2	BldgType		
## 1	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 2	AllPub	FR2	Gtl	Veenker	Feedr	Norm	1Fam		
## 3	AllPub	Inside	Gtl	CollgCr	Norm	Norm	1Fam		
## 4	AllPub	Corner	Gtl	Crawfor	Norm	Norm	1Fam		
## 5	AllPub	FR2	Gtl	NoRidge	Norm	Norm	1Fam		
## 6	AllPub	Inside	Gtl	Mitchel	Norm	Norm	1Fam		
##	HouseStyle	OverallQual	OverallCond	YearBuilt	YearRemodAdd	RoofStyle	RoofMatl		
## 1	2Story	7	5	2003	2003	Gable	CompShg		
## 2	1Story	6	8	1976	1976	Gable	CompShg		
## 3	2Story	7	5	2001	2002	Gable	CompShg		
## 4	2Story	7	5	1915	1970	Gable	CompShg		
## 5	2Story	8	5	2000	2000	Gable	CompShg		
## 6	1.5Fin	5	5	1993	1995	Gable	CompShg		
##	Exterior1st	Exterior2nd	MasVnrType	MasVnrArea	ExterQual	ExterCond	Foundation		
## 1	VinylSd	VinylSd	BrkFace	196	Gd	TA	PConc		
## 2	MetalSd	MetalSd	None	0	TA	TA	CBlock		
## 3	VinylSd	VinylSd	BrkFace	162	Gd	TA	PConc		
## 4	Wd Sdng	Wd Shng	None	0	TA	TA	BrkTil		
## 5	VinylSd	VinylSd	BrkFace	350	Gd	TA	PConc		
## 6	VinylSd	VinylSd	None	0	TA	TA	Wood		
##	BsmtQual	BsmtCond	BsmtExposure	BsmtFinType1	BsmtFinSF1	BsmtFinType2			
## 1	Gd	TA	No	GLQ	706	Unf			
## 2	Gd	TA	Gd	ALQ	978	Unf			
## 3	Gd	TA	Mn	GLQ	486	Unf			
## 4	TA	Gd	No	ALQ	216	Unf			
## 5	Gd	TA	Av	GLQ	655	Unf			
## 6	Gd	TA	No	GLQ	732	Unf			
##	BsmtFinSF2	BsmtUnfSF	TotalBsmtSF	Heating	HeatingQC	CentralAir	Electrical		
## 1	0	150	856	GasA	Ex	Y	SBrkr		
## 2	0	284	1262	GasA	Ex	Y	SBrkr		
## 3	0	434	920	GasA	Ex	Y	SBrkr		
## 4	0	540	756	GasA	Gd	Y	SBrkr		
## 5	0	490	1145	GasA	Ex	Y	SBrkr		
## 6	0	64	796	GasA	Ex	Y	SBrkr		
##	X1stFlrSF	X2ndFlrSF	LowQualFinSF	GrLivArea	BsmtFullBath	BsmtHalfBath	FullBath		
## 1	856	854	0	1710	1	0	2		
## 2	1262	0	0	1262	0	1	2		
## 3	920	866	0	1786	1	0	2		
## 4	961	756	0	1717	1	0	1		
## 5	1145	1053	0	2198	1	0	2		
## 6	796	566	0	1362	1	0	1		
##	HalfBath	BedroomAbvGr	KitchenAbvGr	KitchenQual	TotRmsAbvGrd	Functional			
## 1	1	3	1	Gd	8	Typ			
## 2	0	3	1	TA	6	Typ			

## 3	1	3	1	Gd	6	Typ
## 4	0	3	1	Gd	7	Typ
## 5	1	4	1	Gd	9	Typ
## 6	1	1	1	TA	5	Typ

##	Fireplaces	FireplaceQu	GarageType	GarageYrBlt	GarageFinish	GarageCars
## 1	0	<NA>	Attchd	2003	RFn	2
## 2	1	TA	Attchd	1976	RFn	2
## 3	1	TA	Attchd	2001	RFn	2
## 4	1	Gd	Detchd	1998	Unf	3
## 5	1	TA	Attchd	2000	RFn	3
## 6	0	<NA>	Attchd	1993	Unf	2

##	GarageArea	GarageQual	GarageCond	PavedDrive	WoodDeckSF	OpenPorchSF
## 1	548	TA	TA	Y	0	61
## 2	460	TA	TA	Y	298	0
## 3	608	TA	TA	Y	0	42
## 4	642	TA	TA	Y	0	35
## 5	836	TA	TA	Y	192	84
## 6	480	TA	TA	Y	40	30

##	EnclosedPorch	X3SsnPorch	ScreenPorch	PoolArea	PoolQC	Fence	MiscFeature
## 1	0	0	0	0	<NA>	<NA>	<NA>
## 2	0	0	0	0	<NA>	<NA>	<NA>
## 3	0	0	0	0	<NA>	<NA>	<NA>
## 4	272	0	0	0	<NA>	<NA>	<NA>
## 5	0	0	0	0	<NA>	<NA>	<NA>
## 6	0	320	0	0	<NA>	MnPrv	Shed

##	MiscVal	MoSold	YrSold	SaleType	SaleCondition	SalePrice	economico	medio	caro
## 1	0	2	2008	WD	Normal	208500	1	0	0
## 2	0	5	2007	WD	Normal	181500	1	0	0
## 3	0	9	2008	WD	Normal	223500	1	0	0
## 4	0	2	2006	WD	Abnorml	140000	1	0	0
## 5	0	12	2008	WD	Normal	250000	1	0	0
## 6	700	10	2009	WD	Normal	143000	1	0	0

```

##      Id      MSSubClass      MSZoning      LotFrontage
## Min.    :1461  Min.    : 20.00  Length:1459  Min.    : 21.00
## 1st Qu.:1826  1st Qu.: 20.00  Class :character  1st Qu.: 58.00
## Median :2190  Median : 50.00  Mode  :character  Median : 67.00
## Mean   :2190  Mean   : 57.38                      Mean   : 68.58
## 3rd Qu.:2554  3rd Qu.: 70.00                      3rd Qu.: 80.00
## Max.   :2919  Max.   :190.00                      Max.   :200.00
##                                     NA's    :227
##
##      LotArea      Street      Alley      LotShape
## Min.    : 1470  Length:1459  Length:1459  Length:1459
## 1st Qu.: 7391  Class :character  Class :character  Class :character
## Median : 9399  Mode  :character  Mode  :character  Mode  :character
## Mean   : 9819
## 3rd Qu.:11518
## Max.   :56600
##
##      LandContour      Utilities      LotConfig      LandSlope
## Length:1459      Length:1459      Length:1459      Length:1459
## Class :character  Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
##      Neighborhood      Condition1      Condition2      BldgType
## Length:1459      Length:1459      Length:1459      Length:1459
## Class :character  Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
##      HouseStyle      OverallQual      OverallCond      YearBuilt
## Length:1459      Min.    : 1.000  Min.    :1.000  Min.    :1879
## Class :character  1st Qu.: 5.000  1st Qu.:5.000  1st Qu.:1953
## Mode  :character  Median : 6.000  Median :5.000  Median :1973
##                                     Mean   : 6.079  Mean   :5.554  Mean   :1971
##                                     3rd Qu.: 7.000  3rd Qu.:6.000  3rd Qu.:2001
##                                     Max.    :10.000  Max.    :9.000  Max.    :2010
##
##      YearRemodAdd      RoofStyle      RoofMatl      Exterior1st
## Min.    :1950  Length:1459  Length:1459  Length:1459
## 1st Qu.:1963  Class :character  Class :character  Class :character
## Median :1992  Mode  :character  Mode  :character  Mode  :character
## Mean   :1984
## 3rd Qu.:2004
## Max.   :2010
##
##      Exterior2nd      MasVnrType      MasVnrArea      ExterQual
## Length:1459      Length:1459      Min.    : 0.0  Length:1459
## Class :character  Class :character  1st Qu.: 0.0  Class :character
## Mode  :character  Mode :character  Median : 0.0  Mode  :character

```

```

##                               Mean   : 100.7
##                               3rd Qu.: 164.0
##                               Max.   :1290.0
##                               NA's   :15
##      ExterCond      Foundation      BsmtQual      BsmtCond
## Length:1459      Length:1459      Length:1459      Length:1459
## Class :character  Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##
##
## BsmtExposure      BsmtFinType1      BsmtFinSF1      BsmtFinType2
## Length:1459      Length:1459      Min.   : 0.0      Length:1459
## Class :character  Class :character  1st Qu.: 0.0      Class :character
## Mode  :character  Mode  :character  Median : 350.5    Mode  :character
##                               Mean   : 439.2
##                               3rd Qu.: 753.5
##                               Max.   :4010.0
##                               NA's   :1
##      BsmtFinSF2      BsmtUnfSF      TotalBsmtSF      Heating
## Min.   : 0.00      Min.   : 0.0      Min.   : 0      Length:1459
## 1st Qu.: 0.00      1st Qu.: 219.2    1st Qu.: 784      Class :character
## Median : 0.00      Median : 460.0    Median : 988      Mode  :character
## Mean   : 52.62      Mean   : 554.3      Mean  :1046
## 3rd Qu.: 0.00      3rd Qu.: 797.8      3rd Qu.:1305
## Max.   :1526.00      Max.   :2140.0      Max.   :5095
## NA's   :1          NA's   :1          NA's   :1
##      HeatingQC      CentralAir      Electrical      X1stFlrSF
## Length:1459      Length:1459      Length:1459      Min.   : 407.0
## Class :character  Class :character  Class :character  1st Qu.: 873.5
## Mode  :character  Mode  :character  Mode  :character  Median :1079.0
##                               Mean   :1156.5
##                               3rd Qu.:1382.5
##                               Max.   :5095.0
##
##      X2ndFlrSF      LowQualFinSF      GrLivArea      BsmtFullBath
## Min.   : 0      Min.   : 0.000      Min.   : 407      Min.   :0.0000
## 1st Qu.: 0      1st Qu.: 0.000      1st Qu.:1118      1st Qu.:0.0000
## Median : 0      Median : 0.000      Median :1432      Median :0.0000
## Mean   : 326      Mean   : 3.543      Mean   :1486      Mean   :0.4345
## 3rd Qu.: 676      3rd Qu.: 0.000      3rd Qu.:1721      3rd Qu.:1.0000
## Max.   :1862      Max.   :1064.000      Max.   :5095      Max.   :3.0000
##                               NA's   :2
##      BsmtHalfBath      FullBath      HalfBath      BedroomAbvGr
## Min.   :0.0000      Min.   :0.000      Min.   :0.0000      Min.   :0.000
## 1st Qu.:0.0000      1st Qu.:1.000      1st Qu.:0.0000      1st Qu.:2.000
## Median :0.0000      Median :2.000      Median :0.0000      Median :3.000
## Mean   :0.0652      Mean   :1.571      Mean   :0.3777      Mean   :2.854
## 3rd Qu.:0.0000      3rd Qu.:2.000      3rd Qu.:1.0000      3rd Qu.:3.000
## Max.   :2.0000      Max.   :4.000      Max.   :2.0000      Max.   :6.000
## NA's   :2

```

```

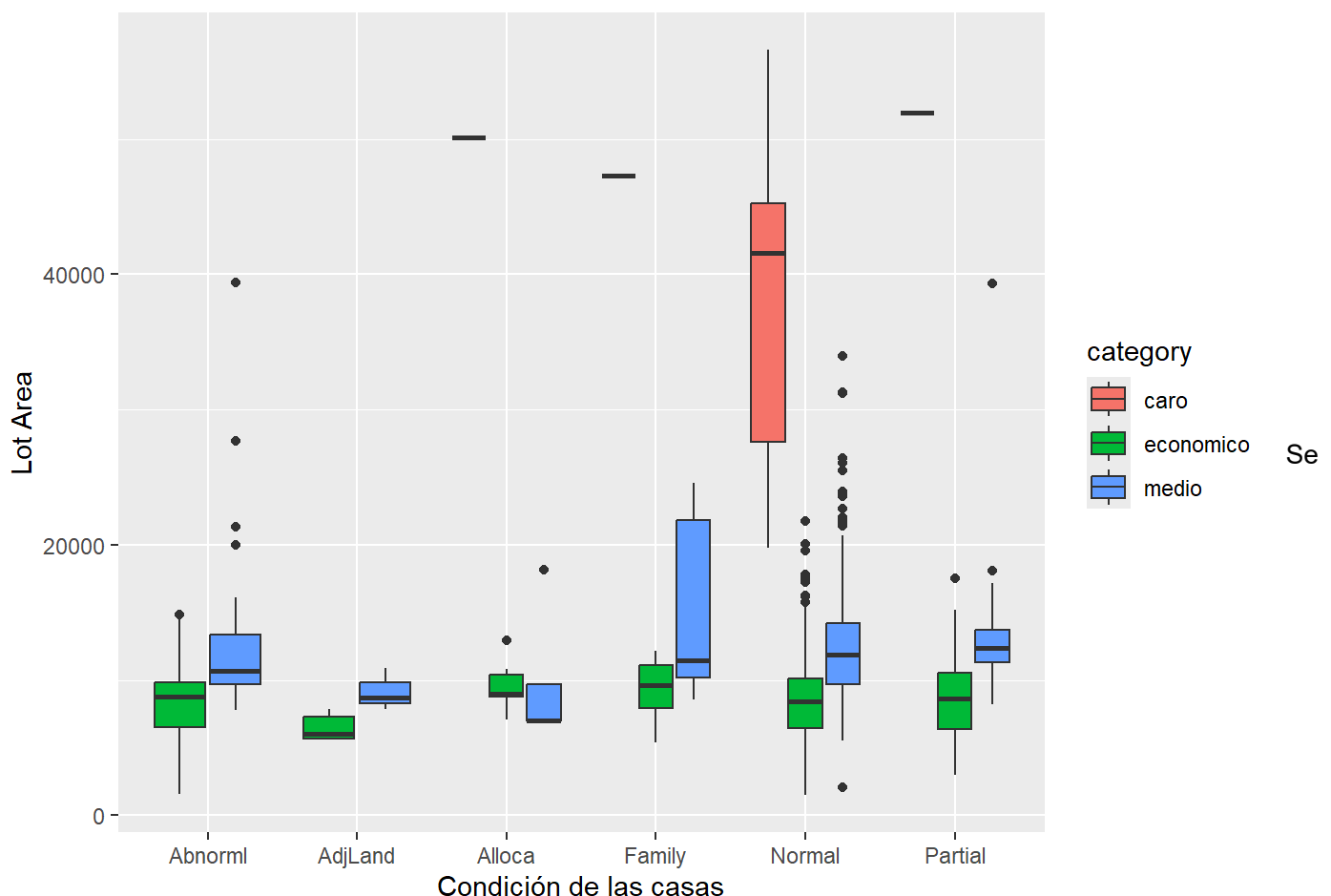
## KitchenAbvGr KitchenQual TotRmsAbvGrd Functional
## Min. :0.000 Length:1459 Min. : 3.000 Length:1459
## 1st Qu.:1.000 Class :character 1st Qu.: 5.000 Class :character
## Median :1.000 Mode :character Median : 6.000 Mode :character
## Mean :1.042 Mean : 6.385
## 3rd Qu.:1.000 3rd Qu.: 7.000
## Max. :2.000 Max. :15.000
##
## Fireplaces FireplaceQu GarageType GarageYrBlt
## Min. :0.0000 Length:1459 Length:1459 Min. :1895
## 1st Qu.:0.0000 Class :character Class :character 1st Qu.:1959
## Median :0.0000 Mode :character Mode :character Median :1979
## Mean :0.5812 Mean :1978
## 3rd Qu.:1.0000 3rd Qu.:2002
## Max. :4.0000 Max. :2207
## NA's :78
## GarageFinish GarageCars GarageArea GarageQual
## Length:1459 Min. :0.000 Min. : 0.0 Length:1459
## Class :character 1st Qu.:1.000 1st Qu.: 318.0 Class :character
## Mode :character Median :2.000 Median : 480.0 Mode :character
## Mean :1.766 Mean : 472.8
## 3rd Qu.:2.000 3rd Qu.: 576.0
## Max. :5.000 Max. :1488.0
## NA's :1 NA's :1
## GarageCond PavedDrive WoodDeckSF OpenPorchSF
## Length:1459 Length:1459 Min. : 0.00 Min. : 0.00
## Class :character Class :character 1st Qu.: 0.00 1st Qu.: 0.00
## Mode :character Mode :character Median : 0.00 Median : 28.00
## Mean : 93.17 Mean : 48.31
## 3rd Qu.: 168.00 3rd Qu.: 72.00
## Max. :1424.00 Max. :742.00
##
## EnclosedPorch X3SsnPorch ScreenPorch PoolArea
## Min. : 0.00 Min. : 0.000 Min. : 0.00 Min. : 0.000
## 1st Qu.: 0.00 1st Qu.: 0.000 1st Qu.: 0.00 1st Qu.: 0.000
## Median : 0.00 Median : 0.000 Median : 0.00 Median : 0.000
## Mean : 24.24 Mean : 1.794 Mean : 17.06 Mean : 1.744
## 3rd Qu.: 0.00 3rd Qu.: 0.000 3rd Qu.: 0.00 3rd Qu.: 0.000
## Max. :1012.00 Max. :360.000 Max. :576.00 Max. :800.000
##
## PoolQC Fence MiscFeature MiscVal
## Length:1459 Length:1459 Length:1459 Min. : 0.00
## Class :character Class :character Class :character 1st Qu.: 0.00
## Mode :character Mode :character Mode :character Median : 0.00
## Mean : 58.17
## 3rd Qu.: 0.00
## Max. :17000.00
##
## MoSold YrSold SaleType SaleCondition
## Min. : 1.000 Min. :2006 Length:1459 Length:1459
## 1st Qu.: 4.000 1st Qu.:2007 Class :character Class :character
## Median : 6.000 Median :2008 Mode :character Mode :character

```

```
## Mean : 6.104 Mean :2008
## 3rd Qu.: 8.000 3rd Qu.:2009
## Max. :12.000 Max. :2010
##
```

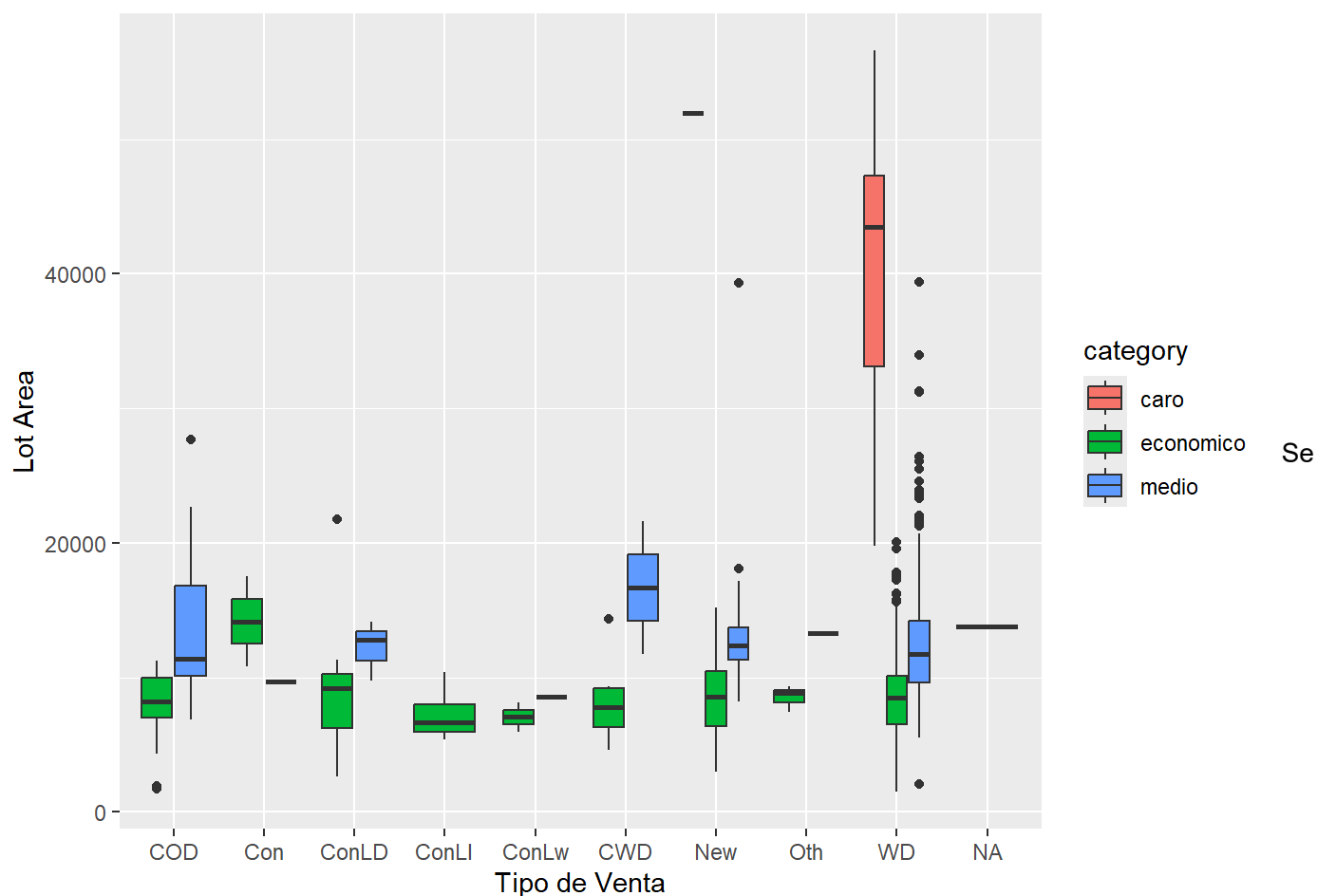
```
##      Id      SalePrice      economico      medio
## Min. :1461 Min. :135751 Min. :0.0000 Min. :0.0000
## 1st Qu.:1826 1st Qu.:168703 1st Qu.:0.0000 1st Qu.:0.0000
## Median :2190 Median :179209 Median :1.0000 Median :0.0000
## Mean :2190 Mean :179184 Mean :0.6923 Mean :0.3009
## 3rd Qu.:2554 3rd Qu.:186789 3rd Qu.:1.0000 3rd Qu.:1.0000
## Max. :2919 Max. :281644 Max. :1.0000 Max. :1.0000
##      caro      category
## Min. :0.000000 Length:1459
## 1st Qu.:0.000000 Class :character
## Median :0.000000 Mode :character
## Mean :0.006854
## 3rd Qu.:0.000000
## Max. :1.000000
```

Se observa que utilizamos para la variable objetivo tres categorías, economico, caro y medio donde primero calculamos el valor de la casa máxima y el mínimo valor de la casa donde luego lo restamos el valor máximo con el mínimo y luego dividirlo entre tres para así diferenciar que casa debería de pertenecer en esas tres variables.

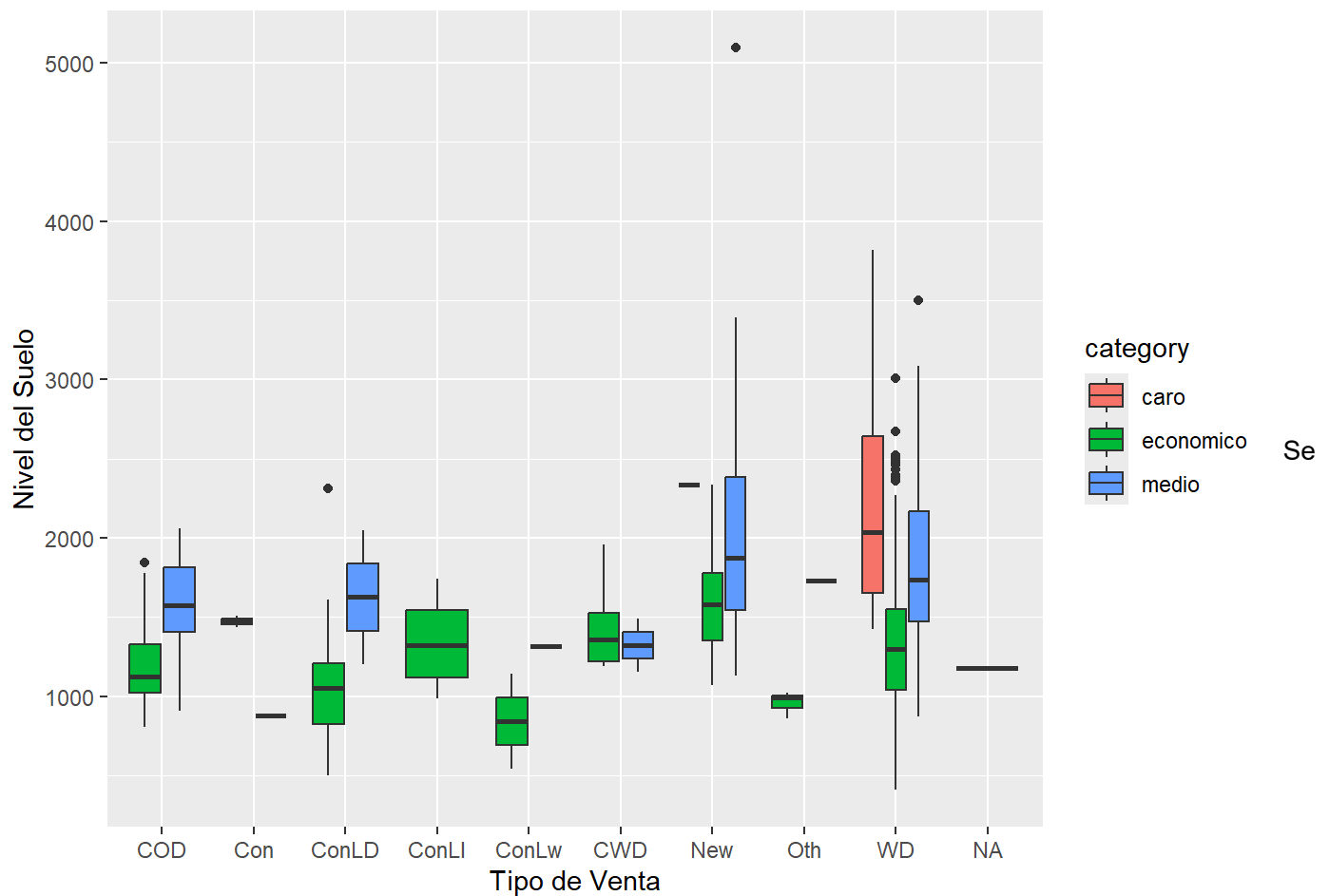


observa que las casas que tienen una condición de venta "Normal" son las más caras. En comparación, otras casas con diferentes condiciones tienden a ser económicas o de bajo costo. Aunque también se observa que la

condición “Family” tiende a presentar un precio medio.



observa que las casas con escritura “WD” presentan un mayor rango de precios, concentrándose en viviendas caras y con áreas superiores a los 30,000 m². Sin embargo, también se identifican varios puntos atípicos en este grupo. Por otro lado, las propiedades con escritura tipo “COD” (escritura emitida por un funcionario judicial) tienden a ubicarse en un rango de entre 0 y 20,000 m², con precios mayormente medios y algunos económicos. Finalmente, las viviendas con tipo de venta “ConLI” (contrato con interés bajo) se asocian exclusivamente a casas económicas, sin presencia en los rangos de precio medio o alto.



observa en este diagrama de caja y bigotes que la mayoría de las casas clasificadas como “caras” forman parte del tipo de venta “WD”, que corresponde a la transferencia mediante escritura. Por otro lado, las categorías “caro” y “medio” se encuentran en un rango similar, aunque “caro” presenta un rango más amplio. También se puede notar que dentro del criterio “WD” existen casos tanto económicos como de precio medio. Por otro lado, las casas vendidas bajo condiciones de menor interés (como “ConLI”) presentan un mayor volumen de viviendas económicas, lo cual resulta coherente con el tipo de financiación ofrecido.